ENGINEERING PORTFOLIO

PROJECT & DESIGN

2026

Nilove Nirupam **Mandal**

MECHANICAL ENGINEERING UNDERGRADUATE

EMAIL
nilovemandal@gmail.com

LOCATION
Jalgaon Jamod, Maharashtra, India.

PHONE
+91 93074 74959

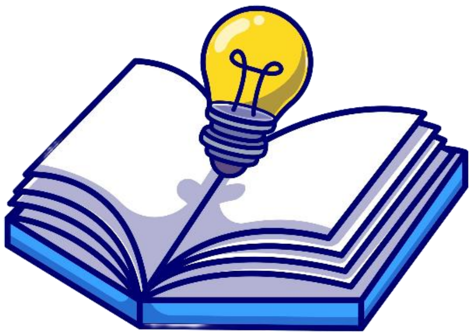
FOCUS
Robotics • Design • IoT

WEBSITE
nilovemandal.site

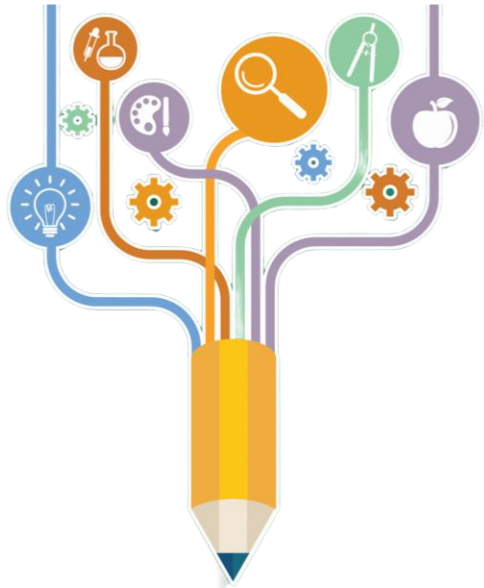
Engineering Portfolio

Nilove Nirupam Mandal

nilovemandal@gmail.com ◆ +91 93074 74959 ◆ Jalgaon Jamod, India.



Welcome, and thank you for taking the time to view my portfolio. The goal of this portfolio is to give you a deeper insight into my experiences and skills I have gained over my recent history



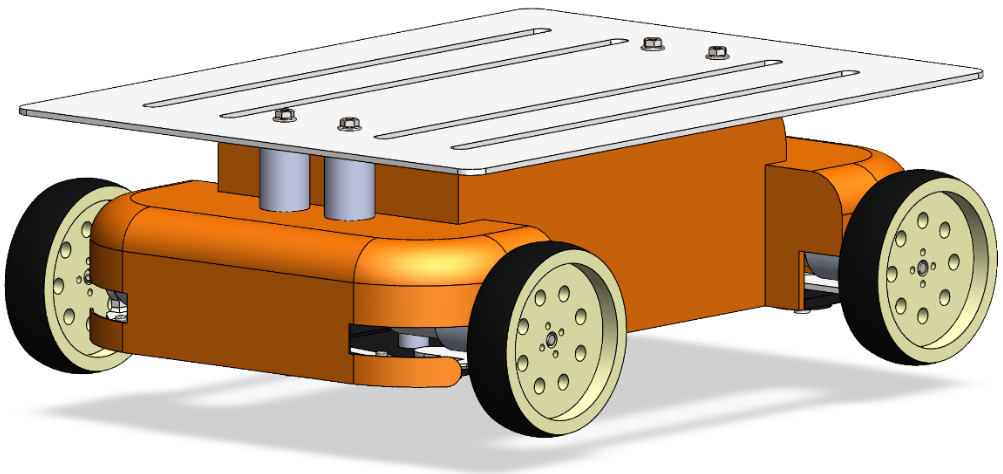
It is my hope that this will allow you to better assess how my skills can be applied to your company.

I would be happy to talk in more detail and can be reached using the contact information on the previous page.

Project Title: Industrial Kuli: Synchronized Crab-Steering AGV

A single-actuator omni-directional vehicle designed for precision logistics in tight industrial spaces.

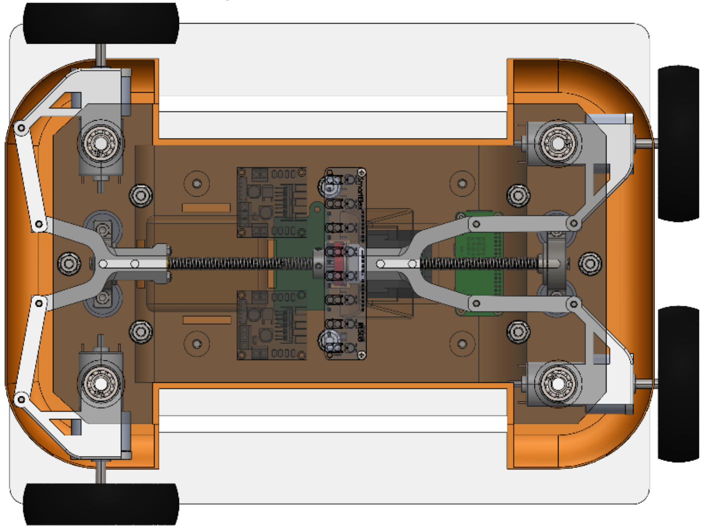
Problem Statement:
Existing industrial Automated Guided Vehicles (AGVs) often require expensive, complex multi-motor systems to achieve crab steering and manoeuvring in narrow warehouse aisles.



Key Achievement:
Successfully achieved mechanical synchronization of 4-wheel steering angles with <2% error variance using a custom dual-nut lead screw drive.

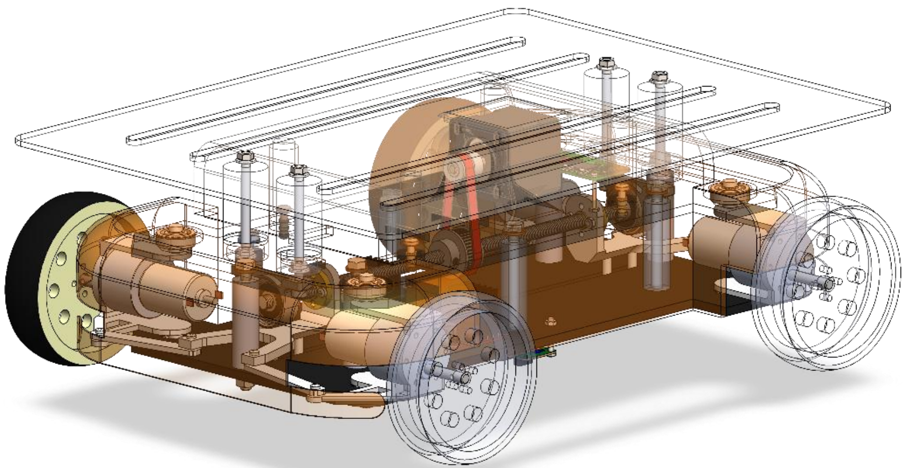
Solution/Impact: The "Kuli" reduces the actuator count by 75% compared to standard 4-wheel independent steering systems, lowering manufacturing costs while maintaining high-precision manoeuvrability for confined industrial environments.

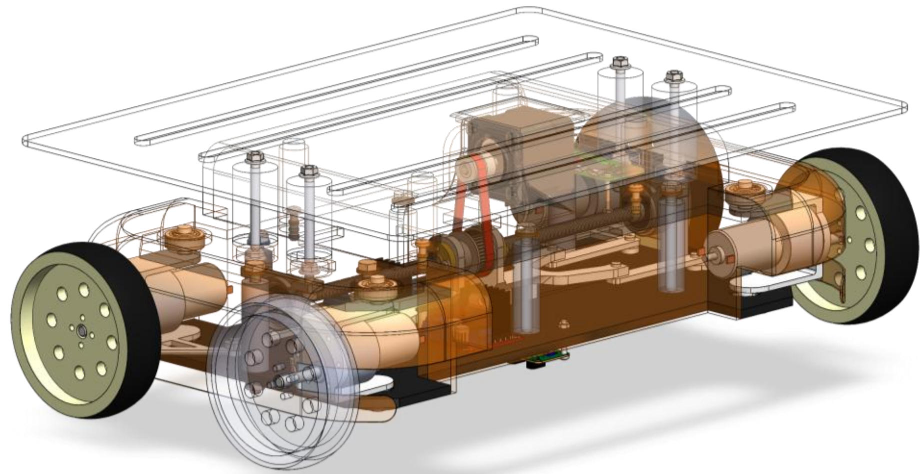
Mechanical Synchronization & Hybrid Navigation



About Features:

- **Single-Actuator Steering**: A central lead screw drives two opposing copper nuts. These nuts push/pull Y-linkages that pivot the wheel hubs, forcing perfect mechanical symmetry in steering angles without complex software feedback loops.
- **Hybrid Guidance**: Combines low-level magnetic tape following for robust path adherence with high-level 2D map planning for route optimization.





Achievement: Validated stable material transport at speeds of 0.3–0.5 m/s while executing lateral (crab) translation.

Project Title: Hootie: *Biomimetic IoT Companion Robot*

An interactive desktop robot blending animatronic biomimicry with IoT connectivity.

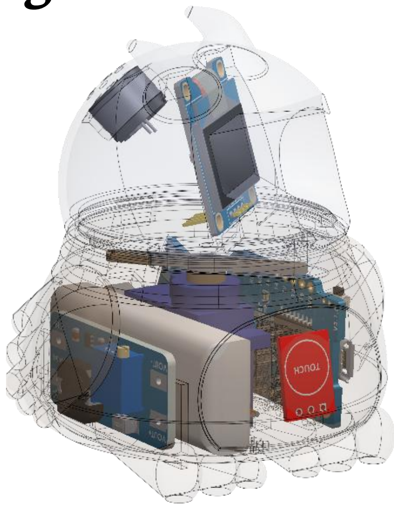
Problem Statement:
Human-Robot Interaction (HRI) often feels sterile; this project aimed to create an emotive, "alive" interface for standard IoT tasks like notifications and timekeeping.

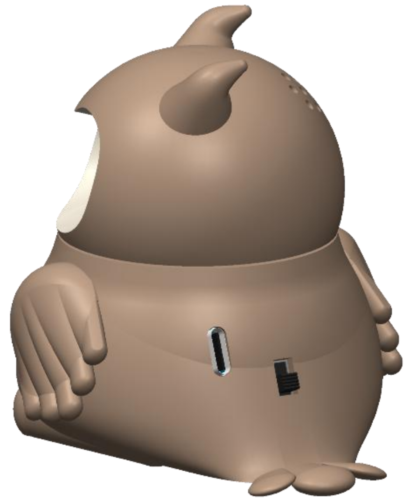
To bridge the gap between user and machine, a capacitive touch sensor was integrated into the robot's anterior shell. This allows Hootie to detect physical affection, such as 'petting.' Upon contact, the system triggers a 'Joy Protocol,' synchronizing happy OLED eye animations, rhythmic servo oscillations (wiggling), and cheerful audio chirps to mimic the response of a living pet. This feature positions Hootie as an engaging 'Desk Buddy' for professionals and an interactive toy for children.

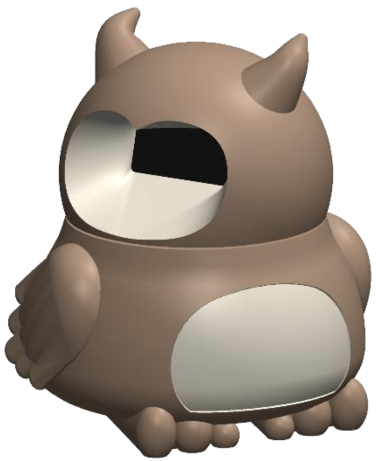
Key Achievement:
Developed a complex C++ state machine with **hardware interrupt handling** for real-time interaction. The system seamlessly transitions between 'Autonomous Life' (random blinking), 'Utility Mode' (Clock), and the new **'Affective Mode,'** where sensor inputs trigger instant audio-visual expressions of joy and excitement.

Solution/Impact: Successfully demonstrated how **multi-modal interaction (Touch-to-Emotion)** significantly deepens emotional engagement with IoT devices. By enabling the robot to respond physically to human touch, Hootie transcends static notification systems, functioning as a therapeutic interactive companion that reduces workspace stress and encourages playful interaction for users of all ages.

Integrated Mechatronics & Web Server Control







About Features:

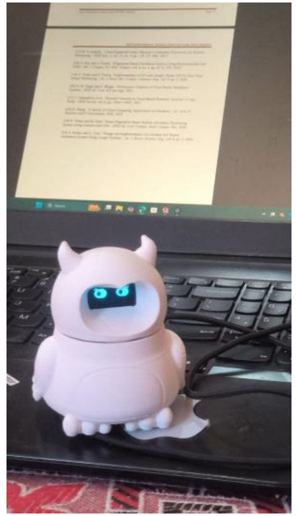
- **Screwless Assembly:** Designed a fully snap-fit enclosure inspired by consumer electronics, utilizing locking head mechanisms to prevent servo strain.
- **IoT "Puppeteer" Mode:** The ESP8266 hosts an asynchronous web server, allowing users to manually control the robot's servo motors and trigger emotion presets (Love, Anger, Confusion) via a smartphone browser.

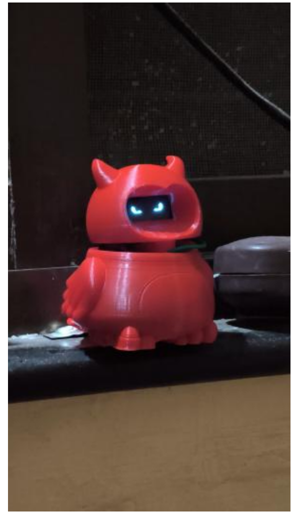
Achievement: Engineered a passive thermal management system that keeps internal temperatures stable even during continuous servo operation.

Owl Project: DFM & Implementation Strategy

Key Engineering Upgrades & Problem Solving:

- **Snap-Fit Redesign:** Early physical prototyping revealed severe structural flaws where the snap-fit joints were too brittle and frequently snapped during assembly. After iterating through 4 different 3D printing phases, the internal geometry was perfected. The final printed shape now features a robust, flawless snap-fit enclosure that holds the internal engines securely.
- **Microcontroller Bottlenecks:** The current architecture relies on a Wemos D1 Mini (ESP8266). Testing revealed that attempting to add more UI elements to the web dashboard causes the microcontroller to crash due to memory limitations.
- **Proposed Processing Upgrade:** To resolve the dashboard stability issues and vastly expand the robot's capabilities, a hardware migration to an ESP32 D1 Mini is currently planned to handle heavier web-server loads and multitasking.





Evolution:

- **Physical Prototyping**: Hootie is currently in an advanced prototype stage. Having successfully finalized the structural 3D prints to prevent part breakage, the physical form factor is completely locked in. The final step remaining in the physical evolution is exterior colouring and finishing.
- **Mechatronic Development**: The robot's internal engines have been tuned to drive physical roleplay and lifelike behaviour, allowing for smooth, natural head panning across a 35 to 145-degree range.
- **Software Maturity**: Now operating on "Version 12 - WEATHER EDITION," the code has evolved from basic reactions to a complex state machine featuring dynamic sleep/wake sequences and real-time clock synchronization via NTP servers.

Advanced Features:

- **Extensive Emotion & Audio Engine:** Features a comprehensive library of over 30 unique moods (e.g., confused, distressed, happy, sleepy) paired with an organic bird audio engine capable of chirps, trills, gruffs, and snores.
- **Live Weather Integration:** Utilizes IP Geolocation and the OpenWeatherMap API to autonomously detect the user's city and fetch real-time ambient temperature, displaying it on the OLED screen during "Clock Mode".
- **Smart Multitasking & Touch:** Employs a custom "Smart Delay" system that continuously polls the capacitive touch sensor even while executing complex eye and servo animations, ensuring the robot never misses a user's pet or touch.
- **Comprehensive Web Dashboard:** A customized HTML interface that allows users to manage Wi-Fi credentials, set daily alarms, configure sleep schedules, and take manual control of the robot's servos and emotions via "Puppet Mode".

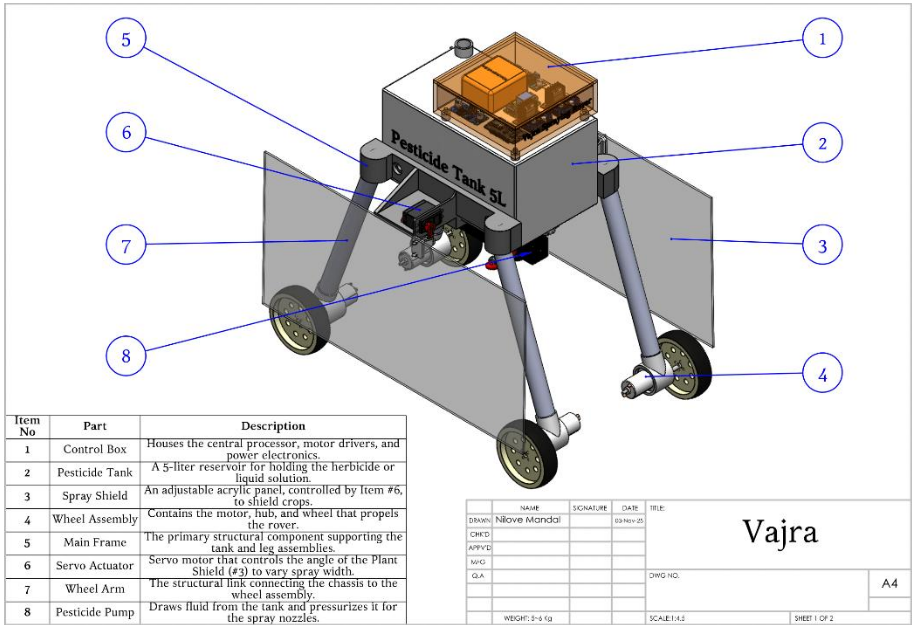
Performance Outcomes:

- **Flawless Affective Response: The touch-based interaction logic performs perfectly. The robot successfully registers affection (petting) and seamlessly translates it into physical wiggling and emotional audio-visual responses.**
- **Robust Autonomous Loop: The core system successfully balances background tasks—such as random autonomous blinking, natural head movements, and trigger-ready alarm logic—without blocking the main thread.**
- **Hardware Stability: With the newly perfected 3D printed chassis, the mechanical engines operate flawlessly without breaking their mounts. Aside from the ESP8266 dashboard limits, the core mechatronic and software systems run with complete stability.**

Project Title: Vajra 2.0: Precision Agrochemical Spraying Platform

A reliability-engineered, teleoperated rover designed to eliminate human exposure to hazardous pesticides.

Problem Statement: Manual pesticide application accounts for significant farmer health risks (inhalation/dermal contact) and environmental waste due to inefficient broadcast spraying.



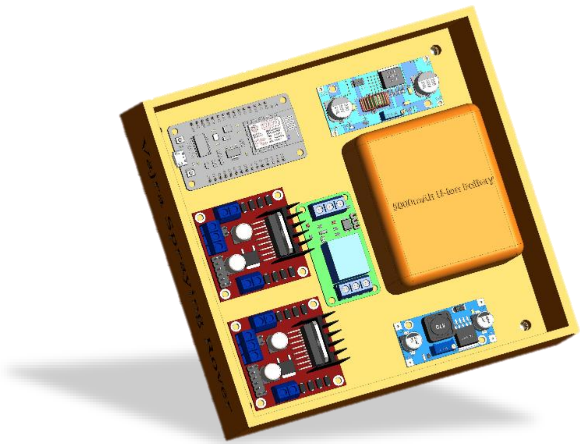
Item No	Part	Description
1	Control Box	Houses the central processor, motor drivers, and power electronics.
2	Pesticide Tank	A 5-liter reservoir for holding the herbicide or liquid solution.
3	Spray Shield	An adjustable acrylic panel, controlled by Item #6, to shield crops.
4	Wheel Assembly	Contains the motor, hub, and wheel that propels the rover.
5	Main Frame	The primary structural component supporting the tank and leg assemblies.
6	Servo Actuator	Servo motor that controls the angle of the Plant Shield (#3) to vary spray width.
7	Wheel Arm	The structural link connecting the chassis to the wheel assembly.
8	Pesticide Pump	Draws fluid from the tank and pressurizes it for the spray nozzles.

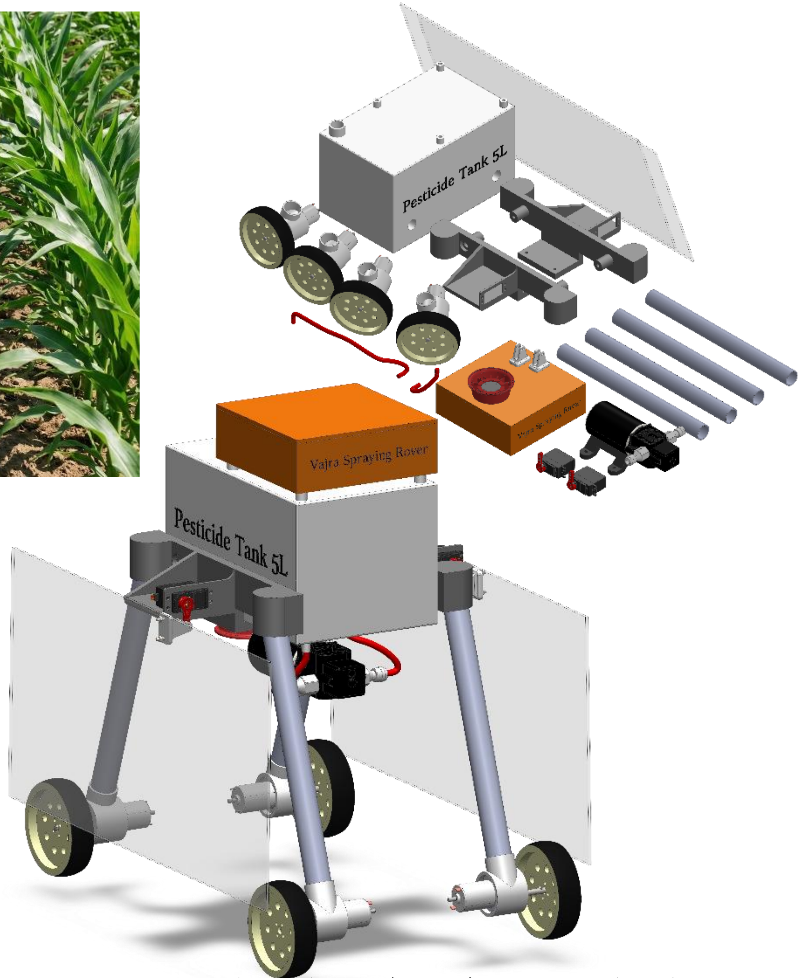
A robust 5-6kg four-wheel-drive rover engineered to carry a 5-liter liquid payload. Unlike typical suspension rovers, Vajra 2.0 utilizes a rigid **Fixed-Arm Chassis** angled at 110° to lower the centre of gravity and counteract the destabilizing "sloshing" forces of the fluid tank.

Key Achievement: Solved the "Liquid Slosh Resonance" instability common in small AGVs by designing a rigid geometry that locks the fluid mass relative to the drive train, ensuring predictable handling on uneven terrain.

Solution/Impact: Acts as a teleoperated proxy, allowing farmers to spray hazardous chemicals from a safe distance, reducing direct exposure risk by 100%.

Drift Control & Modular Architecture





About Features:

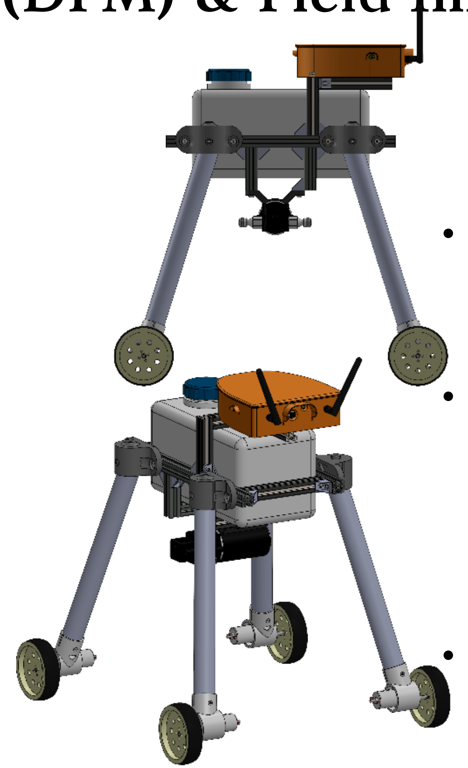
- **Active Spray Shield:** Incorporates a mechanical shielding system to contain spray mist, preventing wind drift and ensuring chemicals land only on the target crop rows.
- **Modular "Brain" Architecture:** Designed with a split-compute architecture. An **ESP32** handles real-time motion control (hard real-time), while the system is pre-wired to accept a **Raspberry Pi 4/5** for future computer vision tasks (crop row detection/weed identification) without redesigning the power train.

Key Achievement: Validated "Design for Reliability" (DFR) principles by prioritizing chassis stiffness and modular repairability over complex suspension systems.

Vajra 2.1: Design for Manufacturing (DFM) & Field Implementation

Key Engineering Upgrades & Problem Solving:

- **Cost-Optimized Chassis & Payload Integration:**
  - *Challenge:* Custom 3D printing the original 5L fluid reservoir and chassis was financially unviable (the tank alone was estimated at ₹18,000) and lacked long-term structural integrity for field use.
  - *Solution:* Transitioned the main chassis to modular **Aluminium Extrusion profiles (₹4,800)**, utilizing heavy-duty nut-and-bolt assemblies alongside redesigned **leg joints (₹1,000/set of 4)** and **motor hubs (₹1,000/set of 4)**. The custom payload was replaced by repurposing a robust, off-the-shelf 5L diesel heater tank (reducing tank cost to **₹2,600**). This comprehensive Design for Manufacturing (DFM) approach drastically improved the rover's rigidity while bringing the entire structural and fluid-holding framework down to a highly optimized budget of just ₹9,400.
- **Compound Stance & Passive Suspension:**
  - *Challenge: Field testing revealed that while a forward inclination prevented forward tipping, the rover was still susceptible to lateral (side-to-side) rollovers on uneven agricultural terrain.*
  - *Solution: Redesigned the leg joint hubs to introduce a **compound inclination angle**—angling the legs 100° from the vertical on both the front/back and left/right axes. The motor hubs were modified to accommodate this new geometry. Additionally, the specific angular design of the connecting pipes between the motor hub and leg joint inherently acts as a **passive suspension system**, flexing naturally to absorb shocks when the 5L tank is fully loaded.*



Advanced Fluid Control & Telemetry:

- *Challenge:* The previous relay-based system only allowed for binary (On/Off) spraying, leading to inefficient pesticide application.
- *Solution:* Integrated a **400W Dual High-Power MOSFET Trigger Switch Drive Module** (5V-36V, 15A continuous). This allows for precise, variable fluid flow control by modulating the pump voltage via PWM signals.
- *Vision Upgrade:* Added an ESP32-CAM module paired with a high-gain 6dBi antenna, enabling stable, real-time First-Person View (FPV) navigation over a local network without relying on external internet infrastructure.

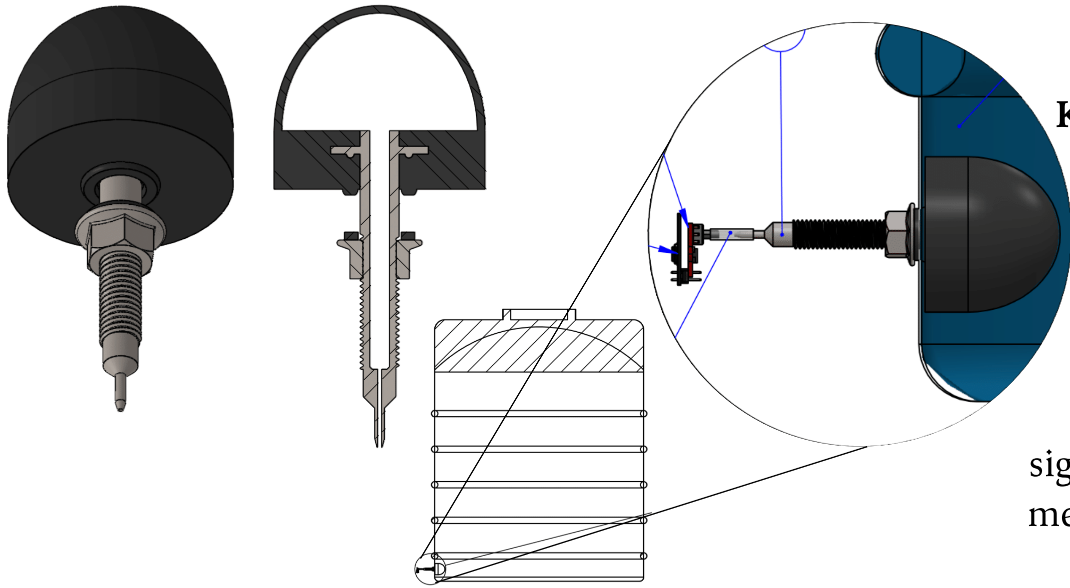
The Evolution: While Vajra 2.0 established the core mechanics, transitioning from a digital CAD environment to a physical, field-ready prototype required significant mechanical and mechatronic overhauls. Vajra 2.1 was engineered with a strict focus on cost-reduction, multi-axis stability, and dynamic fluid control, resulting in a highly capable agricultural rover.

Performance Outcomes: Vajra 2.1 successfully navigates highly uneven terrain without rollover risks, achieving a stable, teleoperated manoeuvring speed of 3 km/hr. The combination of the compound-angle stance, structural cost optimization, and newly implemented PWM variable spray control makes it a highly viable, scalable solution for safe agrochemical application.

Project Title: NR Hydro Gauge: Pneumatic Bladder Level Sensor

A non-contact hydrostatic sensor designed to eliminate electronic failure points in liquid storage tanks.

Problem Statement: Traditional immersion sensors corrode in harsh chemicals, while ultrasonic sensors struggle with foam and condensation in closed tanks.



This system uses a submerged butyl rubber bladder coupled pneumatically to an external pressure sensor. By isolating the electronics outside the tank and using an inline hydrophobic filter/condensate trap, the system prevents liquid ingress and sensor drift caused by humidity.

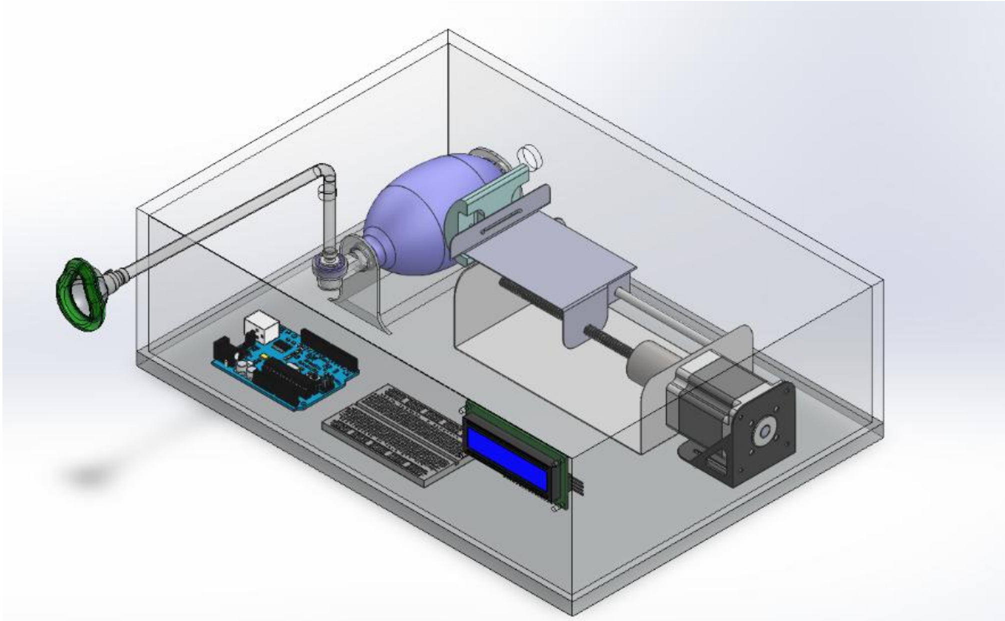
Key Achievement: Engineered a temperature-compensation algorithm that linearizes bladder elasticity offset, significantly improving measurement accuracy over standard hydrostatic probes.

Solution/Impact: Created a retrofittable, low-cost solution that allows for precise level monitoring in opaque or hazardous liquid tanks without exposing sensitive electronics to the fluid.

Smart Breathe: Low-Cost Adaptive Ventilator with Telemetry

Democratizing critical respiratory care with an INR 8,500 IoT-enabled ventilator.

Problem Statement: High-end ventilators (like the Puritan Bennett™ 840) are expensive, complex to operate, and lack portability, creating a critical gap in resource-limited healthcare settings.



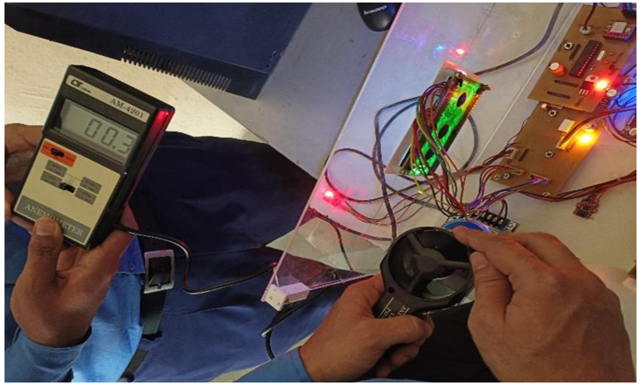
A portable, low-cost mechanical ventilator designed for emergency and home-care use. It automates a silicon Ambu bag (1600ml) using a Nema-17 stepper motor and lead-screw mechanism. Unlike static pump systems, it features an intelligent feedback loop that adjusts airflow based on real-time patient vitals.

Key Achievement: Winner of multiple national innovation awards:

- 1st Prize (Best Innovator): HISET-2024 International Conference at KVVDM Karad.
- 2nd Prize (₹15,000): IIT Ropar’s "IMPACT 3.0" International Event.

Solution/Impact: Validated a cost-effective design (approx. ₹8,500) that provides "Tele-ICU" capabilities, allowing doctors to remotely monitor patient SpO2 and pressure data from anywhere

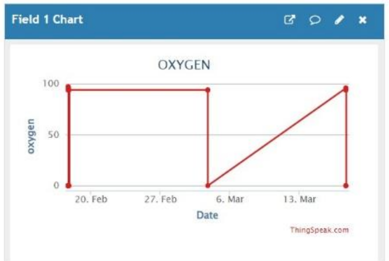
Adaptive Flow Control & IoT Architecture



I. Flow Rate :- 0.3 to 0.4 m/s (90 – 95) oxygen flow
II. Flow Rate :- 0.2 m/s (95 – 100) oxygen level

DATA MONITORING USING IOT:
Health parameters that can be monitored are as follows:

- Oxygen level of patient
- Heart rate of patient
- Air Pressure at delivery side of ventilator
- Temperature of patients body.



About Features:

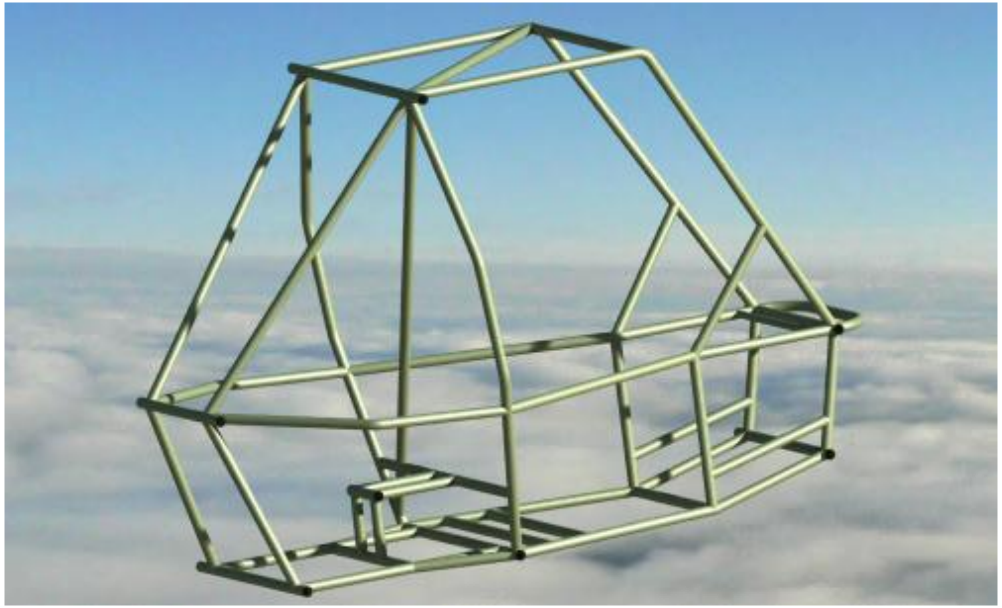
- **Bio-Feedback Loop**: The system reads SpO2 levels via a MAX30100 sensor. If SpO2 drops below 90%, it automatically increases the stroke rate from 10-12 to 15-20 strokes/min to boost oxygenation.
- **Remote Telemetry**: An ESP32 microcontroller transmits patient data (SpO2, Pulse, Air Pressure, Temp) to a cloud dashboard (ThingSpeak) every 60 seconds for trend analysis.

Key Achievement: Validated air discharge of **4.28 Liters/min**, meeting the respiratory requirements for COPD and Asthma support.

Project Title: Team Avibhuj: SAE ATV Roll Cage Design

Optimizing driver safety and chassis stiffness for an all-terrain competition vehicle.

Problem Statement: Designing a vehicle chassis that is lightweight enough for racing yet strong enough to withstand 10ft drop impacts and frontal collisions.

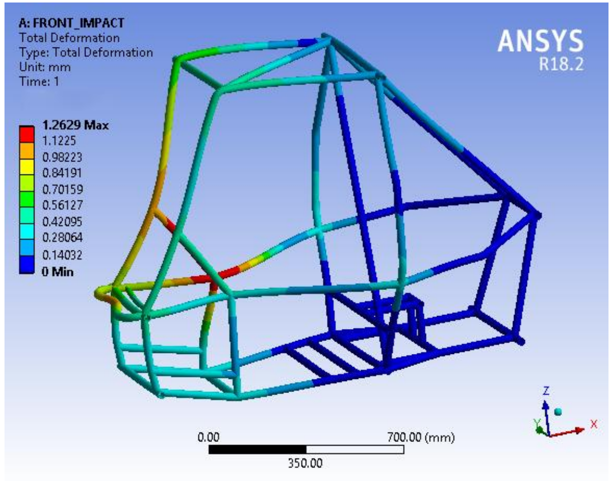


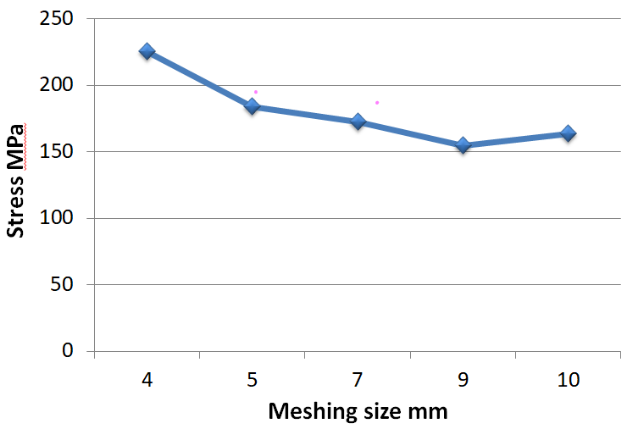
Lead the CAE (Computer-Aided Engineering) evaluation for the roll cage. Utilized CATIA for surfacing and ANSYS Workbench for static and explicit dynamic analysis. We analysed front impact, side impact, and rollover scenarios to ensure compliance with ATVC 2024 Rulebook safety standards.

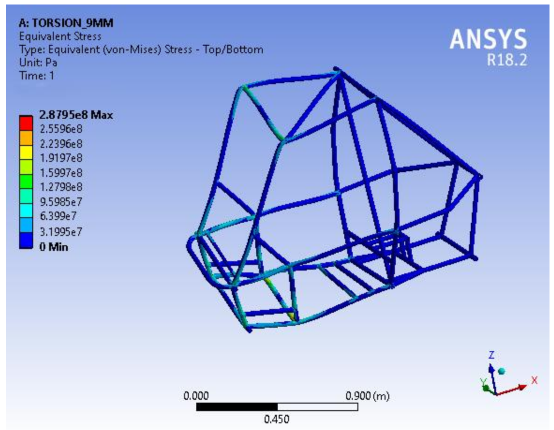
Key Achievement: Achieved a Safety Factor of 2.97 for Front Impact (13,500N load) while maintaining a chassis weight of only 30kg.

Solution/Impact: The optimized AISI 4130 tubular space frame provided superior torsional stiffness (preventing suspension flex transfer) and ensured driver survival space in high-speed crash simulations.

Finite Element Analysis (FEA) & Optimization







About the Special Features:

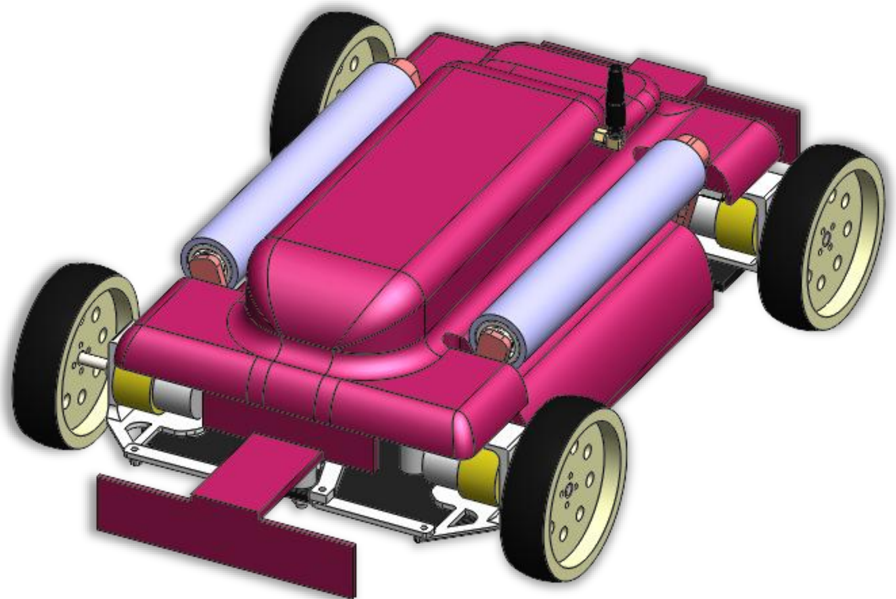
- **Mesh Convergence:** Utilized QUAD/TRIA shell elements for 2D meshing to balance computational efficiency with result accuracy.
- **Modal Analysis:** Calculated natural frequencies (1st mode at 45.4 Hz) to ensure the frame would not resonate with the engine vibration frequency (15-25 Hz), preventing structural fatigue.

Key Achievement: Verified Torsional Stiffness > 3.5 kNm/degree, ensuring the suspension works effectively without chassis twisting.

Project Title: Trinetra V-Clean: Autonomous Solar Panel Cleaning Robot

A vision-guided robotic solution designed to restore peak efficiency to solar arrays using novel multidirectional manoeuvrability.

Problem Statement: Accumulation of dust and bird droppings on solar panels can reduce energy output by up to 25%. Existing cleaning robots often struggle to navigate the gaps and irregular layouts of complex solar farms without manual repositioning.

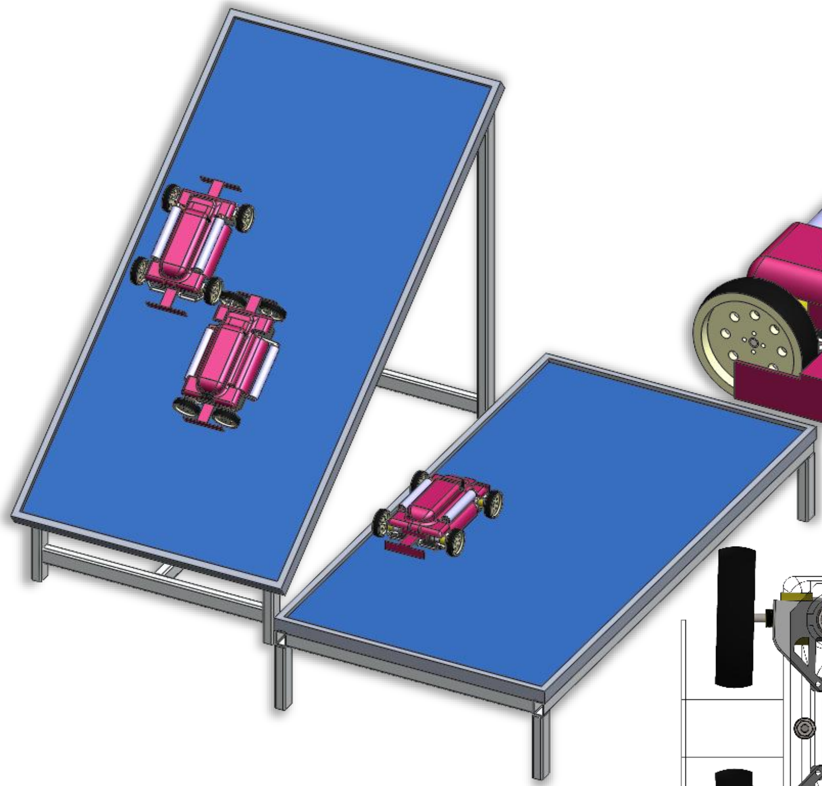


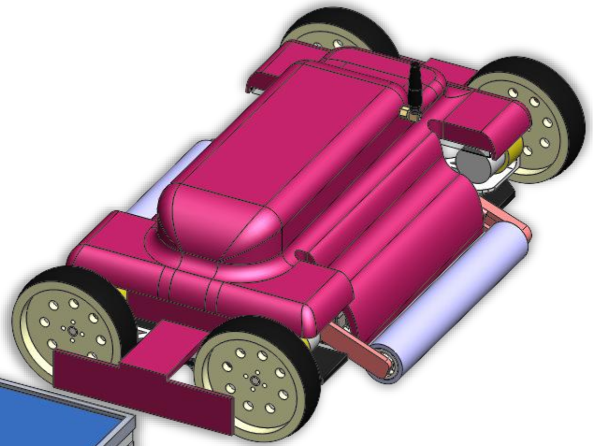
A conceptual design for an autonomous maintenance robot featuring an innovative **"Pivoting Wheel" drive system**. Unlike standard tracked robots that skid-steer, this design allows for holonomic-style movement, enabling the robot to rotate 90 degrees on the spot to transition between adjacent panels in tight or complex array layouts.

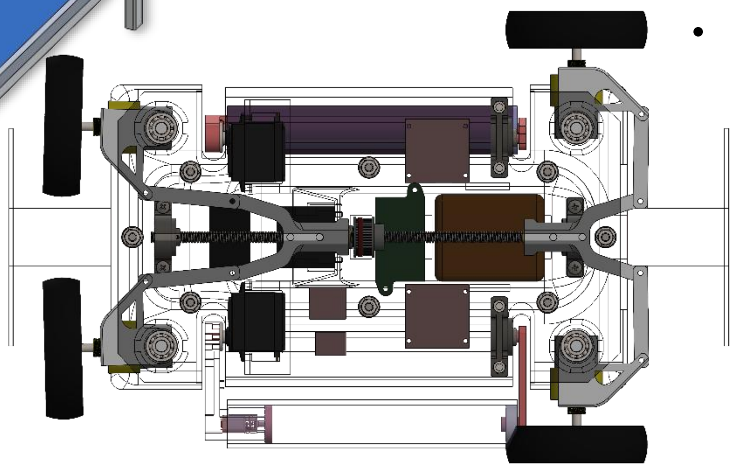
Key Achievement: Designed a unique pivoting drive train mechanism that decouples wheel orientation from chassis direction, allowing for superior manoeuvrability on narrow panel edges..

Solution/Impact: Automating the cleaning process ensures panels operate at maximum capacity, significantly improving the ROI of solar installations while removing humans from dangerous rooftop environments.

Special Features Text: Vision-Based Navigation & Omni-Directional Mobility







About Features:

- **Pivoting Wheel Drive:** The core mechanical innovation is a linkage system that pivots the wheel hubs, allowing the robot to "crab walk" or turn with zero radius. This minimizes the risk of the robot falling into the gaps between panels during edge transitions.
- **Vision Guidance System:** Designed to utilize camera-based edge detection to identify the boundaries of solar panels and detect "hotspots" of dirt, ensuring targeted cleaning rather than random patterns.

Key Achievement: conceptualized a "Complex Array Navigation" logic that allows the robot to clean non-rectangular or segmented solar installations with precision.

Project Title: IoT-Enabled Precision Seeding Unit

Optimizing crop yield potential by strictly enforcing soil-moisture conditions before seeding.

Problem Statement: Sowing seeds in soil with inadequate moisture leads to poor germination rates and wasted resources.

An automated seeding rig that acts as a "Gatekeeper" for planting. It inserts a moisture probe into the soil; if the moisture level is below the threshold, it halts operation and alerts the farmer via Wi-Fi. If levels are optimal, it proceeds to sow seeds in batches.

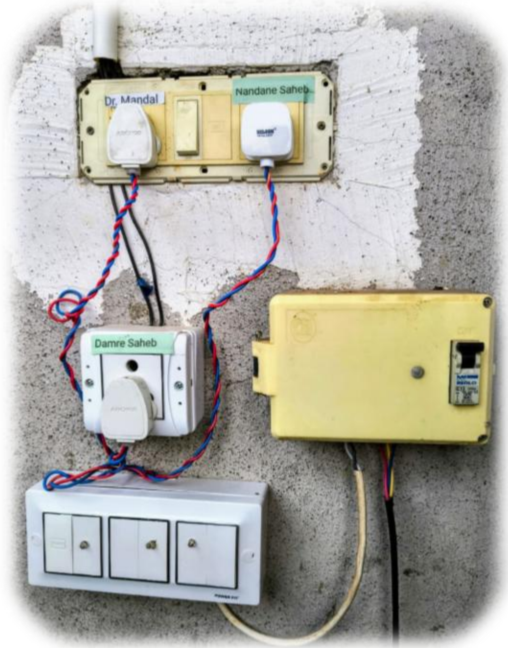
Key Achievement: Programmed a "Batch-Check" algorithm that re-verifies soil conditions after every 10 seeds, ensuring field heterogeneity doesn't compromise the planting batch.

Solution/Impact: Prevents "dead seeds" (failed germination) by ensuring 100% of planted crops are introduced to a viable growth environment.

Project Title: Smart AC Phase Switcher & Load Balancer

An automated arbitration system for fair power distribution in shared-resource environments.

Problem Statement:
In shared housing or multi-meter setups, shared loads (like water pumps) often cause disputes over whose electricity supply is being used, or fail to run if one phase is down.



A solid-state switching unit that monitors AC supply availability from multiple households. It automatically routes the load (pump) to the appropriate power source based on pre-defined priority logic and voltage health.

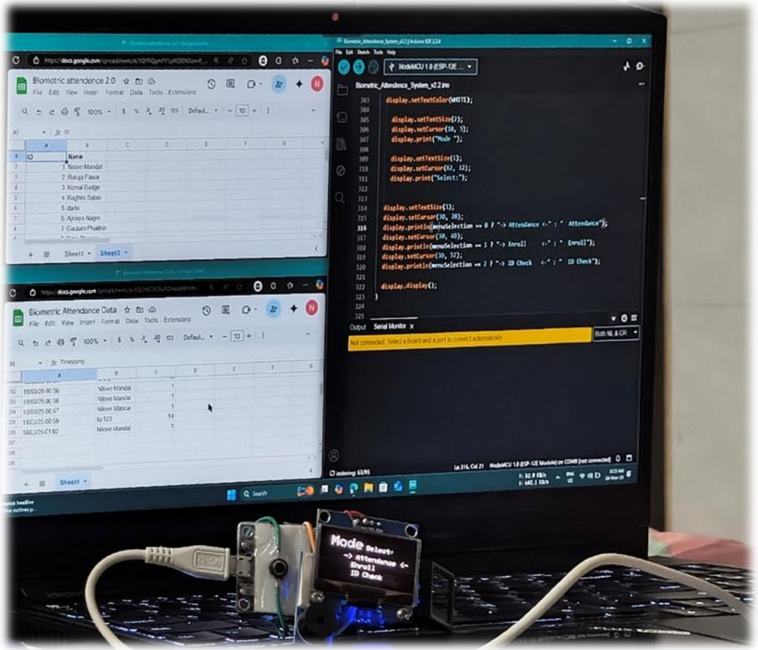
Key Achievement:
Successfully deployed and verified the system in a live multi-house setup, handling inductive loads (pumps) without contact welding or logic failure.

Solution/Impact: Eliminated manual changeover switches and ensured fair electricity billing by automating the source selection based on usage quotas or availability.

Project Title: Wi-Fi-Enabled Biometric Attendance System with Google Sheets Integration

A Wi-Fi-enabled fingerprint logger that eliminates manual data entry by syncing directly to Google Sheets.

Problem Statement: Traditional attendance systems are either offline (requiring manual data transfer) or expensive proprietary solutions that don't integrate easily with common office tools.

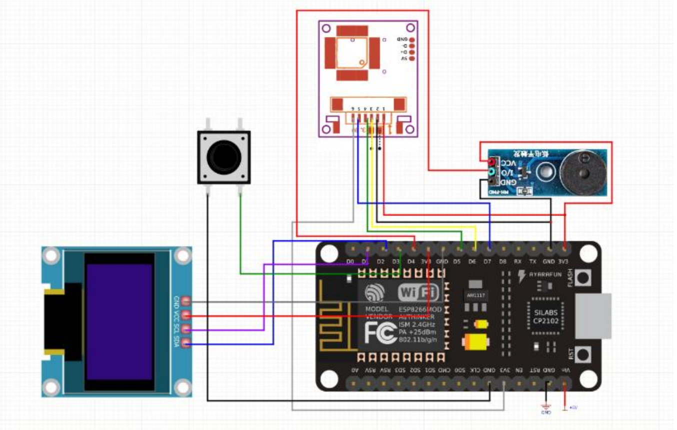


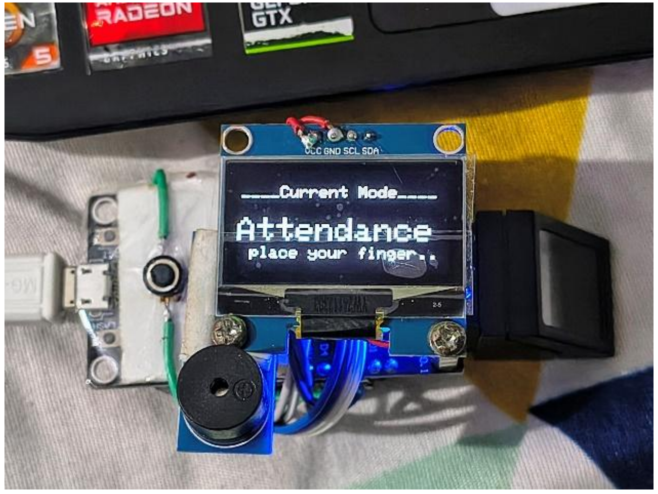
This device scans a user's fingerprint, verifies their identity locally, and immediately pushes the timestamp and user ID to a cloud-based Google Sheet via Wi-Fi. It bridges the gap between physical hardware security and cloud data management.

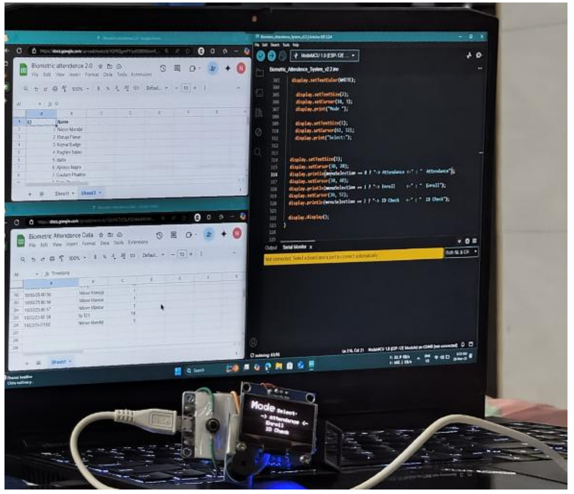
Key Achievement: Hacked the Google Sheets API using Google Apps Script to allow an ESP8266 microcontroller to append rows directly to a secure spreadsheet without an intermediate computer.

Solution/Impact: automated the administrative task of time-logging, reducing payroll processing errors and eliminating "buddy punching" through biometric verification.

IoT Architecture & Data Security







About Features:

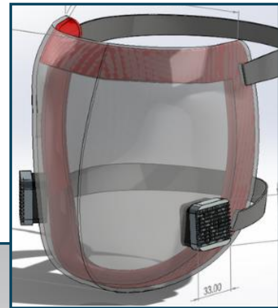
- **Direct-to-Cloud Logging:** Unlike systems that save to an SD card, this system uses an HTTPS request to trigger a script that populates the spreadsheet in real-time.
- **Feedback System:** Features an OLED display and buzzer to give immediate user feedback (Success/Fail) based on server response codes.

Key Achievement: Reduced attendance logging time to <2 seconds per user with 99% reliable cloud sync.

Project Title: Optically Optimized Anti-Fog Face Shield

Redesigning PPE for extended wear comfort and optical clarity in high-humidity medical environments.

Problem Statement: Standard medical face shields suffer from rapid fogging and optical distortion, causing eye strain and reducing visibility for healthcare workers during critical procedures.



A research-driven design project focused on improving the ergonomics and optical properties of protective equipment. The design features a calculated curvature to minimize visual parallax and integrates ventilation channels to promote airflow, reducing fog buildup.

Key Achievement: Conducted comparative research on material coatings and shield geometry to propose a design that maintains 95%+ visible light transmission while mitigating breath condensation.

Solution/Impact: Addressed the "tunnel vision" and fogging issues common in low-cost PPE, providing a design that remains clear during extended periods of physical exertion.

THANK YOU

I look forward to discussing how my experience in mechanical design and robotics can contribute to your organization's goals.

Get in Touch

EMAIL ME

nilovemandal@gmail.com

CALL ME

+91 93074 74959

WEBSITE

nilovemandal.site